



STRIKEQUEST: CARROM LEGENDS

Sprint Planning and Timeline

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PROJECT OVERVIEW

StrikeQuest: Carrom Legends is a realistic 2D physics-based turn-based Carrom board game developed in Unity Engine using C#. The game delivers an authentic Carrom experience with smooth striker mechanics, realistic coin collisions, proper physics, and traditional rules including the Queen covering system. This project is a local 2-player game that focuses on accurate physics simulation, clean code architecture, intuitive controls, and engaging gameplay. The main objective is to pocket all your coins before the opponent while following proper scoring rules.

PROJECT OBJECTIVES

- Develop a realistic 2D physics-based carrom board game
- Implement smooth striker mechanics and aiming system
- Create accurate coin collision and physics simulation
- Implement traditional carrom rules including Queen covering system
- Design attractive board UI with score panels and turn indicators
- Develop clean, modular code architecture using C#
- Deliver a fully functional 2-player carrom game

SPRINT PLANNING

Sprint 1: Asset Creation & Modeling

Objective: Create all game assets and sprites required for gameplay

Tasks:

- Design carrom board sprite
- Create striker sprite
- Design black coins, white coins, and red Queen sprites
- Create corner pocket graphics
- Prepare UI element assets
- Organize sprite folder structure

Deliverable: All game sprites and assets completed

Sprint 2: Physics Setup & Board Configuration

Objective: Configure Unity 2D physics and board layout

Tasks:

- Create carrom board with proper dimensions
- Set up CircleCollider2D for coins
- Configure Rigidbody2D physics properties
- Create physics materials for friction and bounce
- Set up pocket trigger zones
- Configure board surface collision detection

Deliverable: Physics system fully configured and tested

Sprint 3: Striker Control & Aiming System

Objective: Implement striker mechanics and input handling

Tasks:

- Create StrikerController.cs script
- Implement mouse drag aiming system
- Add power meter visualization
- Create aiming line indicator
- Implement force calculation and direction
- Add visual feedback for aiming

Deliverable: Fully functional striker control system

Sprint 4: Game Logic & Scoring System

Objective: Implement core game mechanics and rules

Tasks:

- Create GameManager.cs for game state management
- Implement turn-based system (Player 1 vs Player 2)
- Create ScoreManager.cs for score tracking
- Implement Queen covering logic
- Create pocket detection using triggers
- Implement coin removal and score calculation
- Set up win/lose conditions

Deliverable: Core gameplay mechanics fully functional

Sprint 5: UI Development & Game Flow

Objective: Create user interface and complete game flow

Tasks:

- Create UIManager.cs script
- Design main menu screen
- Create game scene UI (score panels, turn indicator)
- Design power meter and aiming line UI
- Create game-over screen with winner display
- Implement pause menu
- Connect all UI elements to game logic

Deliverable: Complete UI and game flow implemented

Sprint 6: Audio, Polish & Testing

Objective: Add sound effects, optimize, and finalize the game

Tasks:

- Create or integrate coin hit sound effects
- Add pocket sound effects
- Create striker shoot sound effect
- Implement AudioManager.cs for sound management
- Add visual effects (particle system for pockets)
- Perform gameplay testing and bug fixing
- Optimize physics and performance
- Polish overall game feel and responsiveness

Deliverable: Final playable version of StrikeQuest: Carrom Legends

SPRINT TIMELINE

Session	Sprint	Activity
Class 1	Sprint 1	Asset Creation & Modeling
Lab 1	Sprint 2	Physics Setup & Board Configuration
Class 2	Sprint 3	Striker Control & Aiming System
Lab 2	Sprint 4	Game Logic & Scoring System
Class 3	Sprint 5	UI Development & Game Flow
Class 4	Sprint 6	Audio, Polish & Testing

PROJECT DURATION

Total Project Duration:	2 Weeks
Total Sprints:	6
Methodology:	Agile Scrum

EXPECTED OUTCOME

At the end of the development cycle, StrikeQuest: Carrom Legends will be a fully functional 2-player carrom board game featuring realistic physics simulation, smooth striker mechanics, traditional carrom rules with Queen covering system, engaging gameplay, polished UI, sound effects, and optimized performance.